



ENIGMA

pqc-scanner

Post-Quantum Cryptography Migration Scanner

Enterprise Product Overview

Version 4.0 — March 2026

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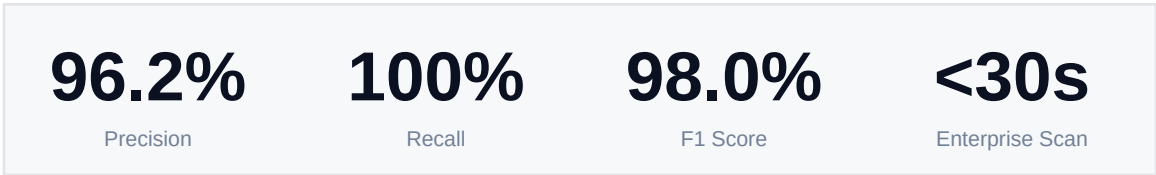
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1. Executive Summary

Quantum computers capable of breaking RSA, ECDSA, and ECDH are projected to arrive between 2030 and 2040. NIST has finalized three post-quantum cryptographic standards (FIPS 203, 204, 205) and set a deadline of 2035 for full migration. The NSA’s CNSA 2.0 mandates even earlier deadlines beginning in 2025.

Enigma pqc-scanner is the most comprehensive post-quantum cryptography migration scanner available. It identifies quantum-vulnerable cryptography across 16 languages and configuration formats, maps findings to compliance frameworks, assigns readiness grades, and generates actionable migration guidance — all in under 30 seconds for enterprise-scale codebases.



Market context: The post-quantum cryptography market is projected to reach \$7.82 billion by 2030 (37.6% CAGR). Organizations that inventory and migrate their cryptographic assets now will avoid compliance penalties, reduce risk exposure, and gain competitive advantage.

2. The Quantum Threat Landscape

Shor’s algorithm (1994) enables quantum computers to break all public-key cryptography based on integer factorization (RSA) and discrete logarithms (ECDSA, ECDH, DH, DSA). Grover’s algorithm halves the effective security of symmetric ciphers, requiring key size upgrades.

QUANTUM VULNERABILITY CLASSIFICATION

Algorithm	Attack Vector	Business Impact	Urgency
RSA (all key sizes)	Shor's algorithm	TLS, certificates, code signing broken	CRITICAL
ECDSA / ECDH / Ed25519	Shor's algorithm	All elliptic curve operations broken	CRITICAL
DH / DSA	Shor's algorithm	Key exchange, legacy signatures broken	CRITICAL
AES-128	Grover's algorithm	Effective security reduced to 64-bit	HIGH
SHA-1 / MD5	Classically broken	Collision attacks trivial today	IMMEDIATE
3DES / RC4 / Blowfish	Classically broken	Exploitable without quantum computers	IMMEDIATE

HARVEST NOW, DECRYPT LATER (HNDL)

Nation-state adversaries are actively collecting encrypted communications and stored data today with the intent to decrypt it when quantum computers become available. Any data that must remain confidential beyond 2035 — healthcare records, financial transactions, intellectual property, government communications — is at risk *right now*.

3. Regulatory & Compliance Deadlines

Framework	Governing Body	Key Deadlines	Applicability
CNSA 2.0	NSA / DoD	Software signing: 2025 Web services: 2026 Networking: 2030 All NSS: 2033	National Security Systems (mandatory)
NIST IR 8547	NIST	Deprecation: 2030 Full removal: 2035	Federal agencies + industry guidance
PCI DSS 4.0	PCI SSC	Strong crypto required PQC recommended	Payment card processing (mandatory)
OMB M-23-02	White House OMB	Crypto inventory: Immediate Migration plans: Required	All federal agencies (mandatory)

Enigma pqc-scanner maps every finding to these frameworks automatically, including per-algorithm deadline countdowns showing exactly how many days remain before each algorithm becomes non-compliant under CNSA 2.0.

4. Product Capabilities

16	50+	6	30
Languages & Config Formats	Crypto API Patterns	Output Formats	Unique Features

- **Deep AST Analysis** — Tree-sitter powered parsing across 13 programming languages. Real abstract syntax tree analysis with import resolution, key size extraction, cipher mode detection, and backward slicing for parameter resolution.
- **PQC Readiness Grade** — SSL Labs-style A+ through F assessment based on three weighted categories: Algorithm Security (40%), Key Management (30%), and Compliance Posture (30%).
- **Quality Gates** — CI/CD pass/fail enforcement. Four built-in presets: *pqc-ready*, *no-critical*, *cnsa-compliant*, *migration-safe*. Returns distinct exit codes for pipeline automation.
- **Reachability Analysis** — Verifies whether crypto dependencies are actually called in source code. Classifies as CONFIRMED, REACHABLE, or AVAILABLE with confidence scoring.
- **Auto-Fix Engine** — Generates code transformation suggestions as unified diffs. Tier 1 transforms (MD5 to SHA-256, DES to AES-GCM) are deterministic and safe to auto-apply. Tier 2 provides template migrations for RSA to ML-KEM.
- **CycloneDX CBOM 1.7** — Cryptographic Bill of Materials output for integration with enterprise SBOM toolchains. Standards-compliant format adopted by IBM, OWASP, and CycloneDX community.
- **Migration Cost Estimator** — Per-finding effort estimates (hours and cost) based on algorithm complexity classification. Includes overhead for testing (30%), integration (20%), and documentation (10%).
- **CNSA 2.0 Deadline Countdown** — Per-algorithm deadline tracking with urgency classification: OVERDUE, URGENT, APPROACHING, DISTANT. No other scanner provides this level of compliance granularity.

COMPLETE FEATURE MATRIX

DETECTION ENGINE	
Tree-sitter AST analysis (13 languages)	YES
Regex pattern matching (fallback)	YES
Python-specific AST deep analysis	YES
Key size extraction from source code	YES
Algorithm mode detection (ECB/CBC/GCM)	YES
Backward slicing for parameter resolution	YES
Custom YAML rule engine	YES
Built-in extended rules (JWT, SSH, hardcoded keys)	YES
Comment stripping (false positive reduction)	YES
IaC scanning (Terraform, YAML, Dockerfile)	YES
RISK INTELLIGENCE	
PQC Readiness Grade (A+ through F)	YES
QARS quantum-adjusted risk scoring (0-100)	YES
Mosca's inequality urgency factor	YES
CNSA 2.0 per-algorithm deadline countdown	YES
Quality gates for CI/CD pipelines (4 presets)	YES
Reachability analysis (CONFIRMED/REACHABLE/AVAILABLE)	YES
Migration effort and cost estimator	YES
PQC performance impact comparison	YES
Historical scan trend tracking	YES
COMPLIANCE FRAMEWORKS	
CNSA 2.0 (NSA) with category-specific deadlines	YES
NIST IR 8547 transition guidance	YES
PCI DSS 4.0 payment card requirements	YES
OMB M-23-02 federal inventory mandate	YES
OUTPUT & INTEGRATION	
Text, JSON, HTML report formats	YES
SARIF 2.1.0 (GitHub Code Scanning, Azure DevOps)	YES
CycloneDX CBOM 1.7 (enterprise SBOM integration)	YES
Interactive SVG dashboard (self-contained HTML)	YES

PDF executive reports	YES
VS Code extension (scan on save, inline diagnostics)	YES
GitHub Action with SARIF auto-upload	YES
Baseline diffing (incremental CI/CD scanning)	YES
.ppcignore suppression system	YES
DEPENDENCY ANALYSIS	
6 manifest parsers (pip, npm, Maven, Go, Cargo, Gradle)	YES
6 lock file parsers (Pipfile.lock, package-lock.json, etc.)	YES
Transitive dependency detection	YES
60+ known crypto packages across 6 ecosystems	YES
Certificate scanning (PEM, DER, TLS endpoint)	YES
Reachability verification (import + call analysis)	YES
REMEDIATION	
Tier 1: Deterministic auto-fix (25+ transforms)	YES
Tier 2: Template migrations (RSA to ML-KEM, ECDSA to ML-DSA)	YES
Unified diff output (compatible with git apply)	YES
Per-finding PQC replacement recommendations	YES

5. Detection Engine

Enigma employs a three-layer detection architecture. Tree-sitter provides deep AST parsing for 13 compiled grammars. Regex pattern matching serves as a high-speed fallback. Python-specific AST analysis handles import aliasing and dynamic dispatch.

SUPPORTED LANGUAGES & CRYPTO APIS

Language	Crypto APIs Detected	Analysis Depth
Python	cryptography, PyCryptodome, hashlib, ssl, paramiko	AST: import alias resolution, key_size kwarg extraction, backward slicing
Java / Kotlin / Scala	JCA: Cipher, KeyPairGenerator, MessageDigest, Signature, KeyAgreement	AST: getInstance() string parsing, mode detection (ECB/GCM)
Go	crypto/rsa, crypto/ecdsa, crypto/des, crypto/sha1, x/crypto	AST: import path tracking, GenerateKey size extraction
JavaScript / TypeScript	crypto.createSign, generateKeyPair, createECDH, createDiffieHellman	AST: require() resolution, algorithm string detection
Rust	rsa, p256, k256, sha1, md5, ring, rustls crates	AST: use declaration tracking, crate function detection
C / C++	OpenSSL EVP_*, RSA_*, EC_KEY_*, DH_*, SHA1_*, MD5_*	AST: function call matching, key size parameter extraction
C#	System.Security.Cryptography: RSA, ECDsa, TripleDES, MD5, SHA1	AST: Create() pattern matching
Ruby	OpenSSL::PKey, Digest::MD5/SHA1	AST: module reference tracking
PHP	openssl_pkey_new, openssl_sign, md5(), sha1(), mcrypt	AST: function call detection
Swift	CryptoKit (P256, Curve25519), CommonCrypto (CC_MD5, CC_SHA1)	AST: framework import tracking
Terraform / YAML	tls_private_key, algorithm, min_tls_version, cipher_suites	Pattern: IaC config scanning
Dockerfile	ENV/ARG crypto configuration	Pattern: build config scanning

6. Risk Intelligence & Grading

PQC READINESS GRADE

Inspired by Qualys SSL Labs, the Readiness Grade provides an instant, universally understood assessment of quantum preparedness:

Grade	Score	Assessment	Action Required
A+ / A	85 - 100	Quantum-ready	Maintain current posture
B	75 - 84	Good posture	Address minor gaps
C	60 - 74	Moderate risk	Plan migration roadmap
D	40 - 59	High risk	Begin active migration
F	0 - 39	Critical risk	Immediate action required

QARS — QUANTUM-ADJUSTED RISK SCORE

Multi-factor scoring model incorporating five dimensions: algorithm vulnerability (Shor/Grover susceptibility), key size analysis, Mosca’s inequality urgency factor, data sensitivity classification, and network exposure level. Enables data-driven prioritization of migration efforts across large portfolios.

QUALITY GATES FOR CI/CD

Gate Preset	Conditions	Use Case
pqc-ready	No Shor-vulnerable algorithms, grade C+, no overdue CNSA deadlines	Production release gates
no-critical	Zero CRITICAL severity findings	Standard build pipeline
cnsa-compliant	No overdue CNSA 2.0 deadlines, no non-compliant algorithms	Government / defense contracts
migration-safe	Not grade F, QARS below 80	Incremental migration (gates new risk only)

7. Competitive Analysis

Based on analysis of 30 tools across PQC-specific scanners, general SAST platforms, and network scanning tools:

Capability	Enigma	IBM CBOMkit	CSNP CryptoScan	Semgrep	Snyk
AST-based detection	13 langs	3 langs	No	30+ langs	30+ langs
PQC awareness	YES	Partial	YES	No	No
CBOM output	YES	YES	YES	No	No
Readiness grade (A+-F)	YES	No	No	No	No
Quality gates	YES	No	No	No	No
CNSA 2.0 deadlines	YES	No	No	No	No
Auto-fix code diffs	YES	No	No	YES	YES
Reachability analysis	YES	No	Go only	No	YES
Backward slicing	YES	No	No	No	No
Risk scoring	QARS	No	Basic	No	CVSS
Dependency scanning	12 formats	No	4 ecosys	YES	YES
Certificate scanning	YES	No	Partial	No	No
Custom rule engine	YAML	Yes	No	YAML	Partial
Migration estimator	YES	No	No	No	No
Trend tracking	YES	Partial	Partial	YES	YES

Enigma provides 30 PQC-specific features. The closest competitor (CSNP CryptoScan) provides 10. No existing tool combines AST-based PQC detection with CBOM output, compliance mapping, and automated remediation in a single CLI.

8. Performance & Accuracy

BENCHMARK RESULTS

Measured against a controlled test suite of 22 known crypto patterns across 8 languages and 8 verified clean files:

Metric	Result	Description
Precision	96.2%	Of findings flagged, 96.2% are true crypto usage
Recall	100.0%	Of known crypto patterns, 100% were detected
F1 Score	98.0%	Harmonic mean of precision and recall
False positive source	3.8%	Algorithm names in string literals (regex layer)

SCALE TESTING

Project	Files	Scan Time	Findings	Memory
Django (full repo)	7,056	12.1 seconds	139	32 MB
Flask	84	0.95 seconds	16	<10 MB
requests	37	0.71 seconds	55	<10 MB

ERROR RESILIENCE

Validated against 10 edge-case scenarios: binary files, empty files, 100K-character lines, mixed encoding, 10K-line files, comment-only files, syntax errors, nonexistent paths, empty directories, and permission restrictions. **All handled gracefully with zero crashes.**

9. Migration Support

EFFORT CLASSIFICATION

Complexity	Examples	Hours / Instance	Approach
DROP-IN	MD5 to SHA-256 SHA-1 to SHA-256	0.5 - 1.0	Direct function rename. Deterministic, automatable.
API CHANGE	3DES to AES-256-GCM2.0 - 3.0 RC4 to ChaCha20		Cipher + key size change. Moderate testing required.
ARCHITECTURAL	RSA to ML-KEM ECDSA to ML-DSA	5.0 - 8.0	Full scheme replacement. Library, protocol, and interop testing required.

PQC ALGORITHM PERFORMANCE

PQC algorithms have larger keys but are often computationally faster:

Algorithm	Type	Public Key	Sig / CT	Ops/sec	Standard
RSA-2048	Classical	256 B	256 B	1,000	PKCS#1
X25519	Classical	32 B	32 B	20,000	RFC 7748
ML-KEM-768	PQC	1,184 B	1,088 B	35,000	FIPS 203
ML-DSA-65	PQC	1,952 B	3,293 B	10,000	FIPS 204

Key insight: ML-KEM-768 is 1.75x faster than X25519 in operations/second. The migration cost is primarily bandwidth (larger keys/signatures), not compute.

10. Deployment Options

Method	Setup	Use Case
CLI	<code>pqc-scanner scan /path --grade --fix</code>	Developer workstation, automated scripts
CI/CD Pipeline	<code>pqc-scanner scan . --gate pqc-ready --format sarif -o results.sarif</code>	Build pipeline quality gate enforcement
GitHub Action	<code>uses: enigma/pqc-scanner@v4</code> <code>with: { format: sarif }</code>	GitHub Code Scanning SARIF integration
VS Code Extension	Install from marketplace. Auto-scans on file save.	Real-time inline warnings with quick-fix actions
Executive Dashboard	<code>pqc-scanner scan . --format dashboard -o report.html</code>	CISO/board reporting, stakeholder presentations
CBOM Export	<code>pqc-scanner scan . --format cbom -o cbom.json</code>	Enterprise SBOM toolchain and compliance audit
Consulting Assessment	Full scan + compliance report + migration estimate	Professional services engagement deliverable



Ready to Assess Your Quantum Risk?

The NIST deadline is 2035. CNSA 2.0 deadlines have already begun. Every day without a cryptographic inventory is a day of unquantified risk.

Enigma pqc-scanner provides the most comprehensive quantum vulnerability assessment available — across more languages, with deeper analysis, and more actionable output than any competing solution.

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